2023 AMC 10B Problem 16 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 16. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 16

Problem:

Define an upno to be a positive integer of 2 or more digits where the digits are strictly increasing moving left to right. Similarly, define a downno to be a positive integer of 2 or more digits where the digits are strictly decreasing moving left to right. For instance, the number 258 is an upno and 8620 is a downno. Let U equal the total number of upnos and D equal the total number of downnos. What is $|U-D|$?

Answer Choices:

A. 512

B. 10

C. 0

D. 9

E. 511

Solution:

Note that an upno is completely determined by the selection of 2 or more elements from the set $\{1,2,3,4,5,6,7,8,9\}$. Because a set with 9 elements has $2^9=512$ subsets, including the empty set and 9 one-element subsets, there are $512-9-1=502$ upnos, so $U=502$. By similar logic using the set $\{0,1,2,3,4,5,6,7,8,9\}$, there are $2^{10}-10-1=1024-11=1013$ downnos, so $D=1013$. Thus $|U-D|=|502-1013|=(\text{E})\boxed{511}$.

$$\textbf{OR}$$

For every upno, there is a corresponding downno formed by reversing the digits. The downnos that are not created in this way must end in 0 . There are $2^9 - 1 = 511$ non-empty subsets of $\{1, 2, \ldots, 9\}$, each of which corresponds to a unique downno ending in 0 . Thus $|U - D| = |-511| = \text{(E)} \boxed{511}$.